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TREND-BASED MOCK PAPER

GATE Mechanical Engineering

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Mechanical Engineering
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Area of circle radius r :

(Numerical Answer Type — Enter the numerical value)

Q2. [1-MCQ] [1 mark]

Antonym of 'prolific':

- (A) Scarce
- (B) Abundant
- (C) Creative
- (D) Unusual

Section B: Mechanical Engineering

Q3. [1-MCQ] [1 mark]

Derivative of $\sin(x)$:

- (A) $\cos(x)$
- (B) $-\sin(x)$
- (C) $-\cos(x)$
- (D) $\tan(x)$

Q4. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q5. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q6. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q7. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q8. [2-MSQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q9. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q10. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q11. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q12. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q13. [1-NAT] [1 mark]

First law based on:

(Numerical Answer Type — Enter the numerical value)

Q14. [1-MCQ] [1 mark]

Entropy change for reversible process:

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Infinity

Q15. [1-MCQ] [1 mark]

Entropy change of universe for reversible process:

- (A) >0
- (B) $=0$
- (C) <0
- (D) Depends

Q16. [2-MCQ] [2 marks]

Max efficiency of Carnot between 500K and 300K:

- (A) 40%
- (B) 50%
- (C) 60%
- (D) 80%

Q17. [2-MSQ] [2 marks]

$C_p - C_v$ for ideal gas =

- (A) 0
- (B) R
- (C) 2R
- (D) R/2

Q18. [2-NAT] [2 marks]

Isothermal expansion: $V_2=2V_1$, $T=300\text{K}$. Work per mole:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

Air at 27°C, 1 atm compressed to 10 atm isentropically ($\gamma=1.4$):

- (A) 534 K
- (B) 507 K
- (C) 300 K
- (D) 600 K

Q20. [2-NAT] [2 marks]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

(Numerical Answer Type — Enter the numerical value)

Q21. [2-MCQ] [2 marks]

For Otto cycle, efficiency depends on:

- (A) Only compression ratio
 - (B) r and γ
 - (C) Temperature only
 - (D) Pressure only
-

Q22. [1-MCQ] [1 mark]

Bernoulli applies to:

- (A) Viscous
 - (B) Inviscid
 - (C) Compressible
 - (D) All
-

Q23. [1-NAT] [1 mark]

Laminar flow has:

(Numerical Answer Type — Enter the numerical value)

Q24. [1-NAT] [1 mark]

Dynamic viscosity of water at 20C approx:

(Numerical Answer Type — Enter the numerical value)

Q25. [1-NAT] [1 mark]

Laminar pipe flow: Hagen-Poiseuille $h_f =$

(Numerical Answer Type — Enter the numerical value)

Q26. [2-MCQ] [2 marks]

Boundary layer at $x=1m$, $Re_x=5e5$ (Blasius):

- (A) 24.5mm
 - (B) 14.1mm
 - (C) 2.45mm
 - (D) 1.41mm
-

Q27. [2-MCQ] [2 marks]

Pitot tube measures:

- (A) Static pressure
 - (B) Dynamic pressure
 - (C) Total pressure
 - (D) Viscosity
-

Q28. [2-MCQ] [2 marks]

Bernoulli total head =

- (A) $P/\rho + V^2/2$
- (B) $P/\rho + V^2/2 + gz$
- (C) $P + \rho * gz$
- (D) $\rho * V^2/2$

Q29. [2-MCQ] [2 marks]

Venturi $D_1=20\text{cm}$, $D_2=10\text{cm}$, $h=10\text{cm}$ water. Q :

- (A) $0.044 \text{ m}^3/\text{s}$
- (B) $0.022 \text{ m}^3/\text{s}$
- (C) $0.088 \text{ m}^3/\text{s}$
- (D) $0.011 \text{ m}^3/\text{s}$

Q30. [2-MCQ] [2 marks]

Water 5cm pipe at 3 m/s. Re ($\nu=1\text{e-}6$):

- (A) $1.5\text{e}5$
- (B) $1.5\text{e}6$
- (C) $1.5\text{e}4$
- (D) $1.5\text{e}3$

Q31. [1-MCQ] [1 mark]

Pattern volume for 1000 cm^3 casting with 2% shrinkage:

- (A) 1020 cm^3
- (B) 1000 cm^3
- (C) 980 cm^3
- (D) 1010 cm^3

Q32. [2-NAT] [2 marks]

Taylor tool life $VT^n=C$. Speed doubled, life:

(Numerical Answer Type — Enter the numerical value)

Q33. [2-NAT] [2 marks]

Six Sigma defects per million:

(Numerical Answer Type — Enter the numerical value)

Q34. [2-MCQ] [2 marks]

Lathe: workpiece rotates against:

- (A) Static tool
- (B) Rotating tool
- (C) Oscillating
- (D) Abrasive

Q35. [2-MSQ] [2 marks]

Tresca theory for:

- (A) Brittle
- (B) Ductile
- (C) Both
- (D) Neither

Q36. [2-MCQ] [2 marks]

Orthogonal cutting shear angle:

- (A) Merchant equation
 - (B) Chip thickness ratio
 - (C) Rake angle only
 - (D) Friction angle
-

Q37. [2-MCQ] [2 marks]

ECM (Electrochemical Machining) removes metal by:

- (A) Cutting
- (B) Electrochemical dissolution
- (C) Abrasion
- (D) Melting

Q38. [1-MCQ] [1 mark]

Stefan-Boltzmann: $q = ?$

- (A) $<5\sigma$
- (B) $<5\epsilon\sigma$
- (C) $<5\epsilon\sigma_0$
- (D) $<5\epsilon\sigma_b$

Q39. [1-NAT] [1 mark]

Fourier's law $q =$:

(Numerical Answer Type — Enter the numerical value)

Q40. [1-NAT] [1 mark]

$k=0.5$, $L=0.2\text{m}$, $A=10\text{m}^2$, $dT=50\text{K}$. Q :

(Numerical Answer Type — Enter the numerical value)

Q41. [1-MCQ] [1 mark]

$Bi \ll 1$ implies:

- (A) Fin
- (B) Lumped capacitance
- (C) S-B law
- (D) Newton cooling

Q42. [2-MCQ] [2 marks]

Shape factor sphere to large enclosure:

- (A) 1
- (B) $4\pi r^2$
- (C) πr^2
- (D) r

Q43. [2-NAT] [2 marks]

Black body at 1000K emissive power:

(Numerical Answer Type — Enter the numerical value)

Q44. [2-MCQ] [2 marks]

Thermal resistance of plane wall:

- (A) $L/(kA)$
- (B) kA/L
- (C) kL/A
- (D) $A/(kL)$

Q45. [2-MCQ] [2 marks]

Natural convection Nu for vertical plate ($10^4 \leq GrPr \leq 10^9$):

- (A) $0.59(GrPr)^{1/4}$

- (B) $0.68 \cdot (Gr \cdot Pr)^{1/2}$
- (C) $0.027 \cdot (Gr \cdot Pr)^{3/4}$
- (D) $0.85 \cdot (Gr \cdot Pr)^{1/3}$

Q46. [1-MCQ] [1 mark]

Gear ratio is:

- (A) Teeth ratio
- (B) Speed ratio
- (C) Torque ratio
- (D) Power ratio

Q47. [1-NAT] [1 mark]

Cam converts:

(Numerical Answer Type — Enter the numerical value)

Q48. [1-NAT] [1 mark]

4-bar: $s+l$ vs $p+q$. Equality means:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-NAT] [1 mark]

Gyroscopic couple: $I=2$, $\omega_s=3000\text{rpm}$, $\omega_p=20\text{rad/s}$:

(Numerical Answer Type — Enter the numerical value)

Q50. [2-NAT] [2 marks]

Grashof $s+l$ < $p+q$: fixed shortest =

(Numerical Answer Type — Enter the numerical value)

Q51. [2-MCQ] [2 marks]

Governor maintains:

- (A) Constant speed
- (B) Constant torque
- (C) Constant power
- (D) Variable speed

Q52. [2-MCQ] [2 marks]

Belt power: $P = (T_1 - T_2) \cdot V$. $V =$

- (A) Belt linear velocity
- (B) Angular
- (C) Diameter
- (D) Torque

Q53. [2-NAT] [2 marks]

SHM cam follower. Jerk max at:

(Numerical Answer Type — Enter the numerical value)

Q54. [1-MCQ] [1 mark]

SS beam UDL w N/m, span L. Max BM:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL^2/4$

(D) $wL^2/2$

Q55. [2-MCQ] [2 marks]

Cantilever UDL w over L . Deflection:

- (A) $wL^4/(8EI)$
- (B) $wL^4/(3EI)$
- (C) $5wL^4/(384EI)$
- (D) $PL^3/(3EI)$

Q56. [2-MCQ] [2 marks]

Max shear stress in solid shaft:

- (A) $16T/(\pi d^3)$
- (B) $32T/(\pi d^3)$
- (C) $T/J \cdot R$
- (D) $T \cdot d/2$

Q57. [2-NAT] [2 marks]

Fixed-free Euler buckling:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

Stress = Force /:

- (A) Length
- (B) Area
- (C) Volume
- (D) Mass

Q59. [1-MCQ] [1 mark]

Section modulus Z =:

- (A) I/y
- (B) I/G
- (C) M/Z
- (D) $\tau \cdot \rho$

Q60. [1-NAT] [1 mark]

Sand casting uses:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

MIG welding uses:

- (A) Flux
- (B) Gas shield
- (C) Both
- (D) None

Q62. [1-MCQ] [1 mark]

$F = ma$ is:

- (A) Newton 1st
- (B) 2nd
- (C) 3rd
- (D) None

Q63. [1-MCQ] [1 mark]

Equilibrium requires:

- (A) :4cÓ
- (B) :4ÓÓ
- (C) Both
- (D) None

Q64. [1-MCQ] [1 mark]

Shaft designed for:

- (A) Torsion
- (B) Bending
- (C) Combined
- (D) Axial

Q65. [1-MCQ] [1 mark]

Key prevents:

- (A) Sliding
 - (B) Rotation
 - (C) Both
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	A	1
3	1MCQ	A	1
4	1MCQ	B	1
5	1MCQ	A	1
6	1MCQ	B	1
7	1NAT	A	1
8	1NAT	B	1
9	1MCQ	B	1
10	1NAT	A	1
11	1MCQ	A	1
12	1MCQ	D	1
13	1NAT	A	1

14	1MCQ	B	1
15	1NAT	B	1
16	1MCQ	B	1
17	1MCQ	C	1
18	1MCQ	C	1
19	1MCQ	C	1
20	1MCQ	A	1
21	1NAT	A	1
22	1NAT	A	1
23	1NAT	A	1
24	1MCQ	A	1
25	1NAT	C	1
26	1MCQ	B	1
27	1NAT	C	1
28	1MCQ	C	1

29	1MCQ	B	1
30	1MCQ	A	1
31	2MCQ	B	2
32	2MCQ	A	2
33	2MCQ	A	2
34	2NAT	A	2
35	2NAT	A	2
36	2MCQ	B	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2NAT	B	2
40	2NAT	A	2
41	2MSQ	B	2
42	2MCQ	A	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2MCQ	C	2
46	2MCQ	B	2
47	2MCQ	A	2
48	2MSQ	B	2
49	2MSQ	A	2
50	2MCQ	A	2
51	2NAT	A	2
52	2MCQ	A	2
53	2MCQ	A	2
54	2MCQ	B	2
55	2MCQ	A	2
56	2MCQ	B	2
57	2NAT	B	2
58	2MCQ	B	2

59	2NAT	A	2
60	2MCQ	B	2
61	2MCQ	B	2
62	2MCQ	A	2
63	2MCQ	A	2
64	2NAT	C	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: difficult

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Strength of Materials | Topic: Stress | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Strength of Materials | Topic: Bending | Difficulty: easy

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Fluid Mechanics | Topic: Bernoulli | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Fluid Mechanics | Topic: Viscosity | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Manufacturing Processes | Topic: Casting | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Manufacturing Processes | Topic: Welding | Difficulty: easy

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Thermodynamics | Topic: First Law | Difficulty: moderate

Q11.

Answer: A

Solution:

Correct Answer: A

Subject: Thermodynamics | Topic: Entropy | Difficulty: easy

Q12.

Answer: D

Solution:

Correct Answer: D

Subject: Heat Transfer | Topic: Radiation | Difficulty: moderate

Q13.

Answer: A

Solution:

Correct Answer: A

Subject: Heat Transfer | Topic: Conduction | Difficulty: difficult

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Machines | Topic: Gears | Difficulty: moderate

Q15.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Machines | Topic: Cam | Difficulty: moderate

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mechanics | Topic: Dynamics | Difficulty: easy

Q17.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mechanics | Topic: Statics | Difficulty: difficult

Q18.

Answer: C

Solution:

Correct Answer: C

Subject: Machine Design | Topic: Shaft | Difficulty: moderate

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Machine Design | Topic: Keys | Difficulty: difficult

Q20.

Answer: A

Solution:

Pattern = Casting/(1-0.02) = 1000/0.98 = 1020.4 cm³.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q21.

Answer: A

Solution:

mu_{water}(20C) approx 1e-3 Pa.s or 1 cP.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q22.

Answer: A

Solution:

$$Q = k \cdot A \cdot \Delta T / L = 0.5 \cdot 10 \cdot 50 / 0.2 = 1250 \text{ W.}$$

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q23.

Answer: A

Solution:

$$\text{Laminar: } h_f = 32 \cdot \mu \cdot L \cdot V / (\rho \cdot g \cdot D^2).$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q24.

Answer: A

Solution:

$$\text{SS UDL: max BM at center} = wL^2/8.$$

Subject: SOM | Topic: SOM | Difficulty: easy

Q25.

Answer: C

Solution:

$s+l = p+q$: change point mechanism. All four links can become cranks.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: easy

Q26.

Answer: B

Solution:

$Bi < 0.1$: lumped capacitance valid.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q27.

Answer: C

Solution:

$$C = I \cdot \omega_s \cdot \omega_p = 2 \cdot 314.16 \cdot 20 = 12566 \text{ Nm.}$$

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: difficult

Q28.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q30.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q31.

Answer: B

Solution:

$$\Delta_{99} = 5x/\sqrt{\text{Re}} = 5 \cdot 1000/\sqrt{5e5} = 7.07 \text{ mm. } \Delta_{99\%} \approx 7 \text{ mm. Close to B.}$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q32.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: A

Solution:

$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q34.

Answer: A

Solution:

$T_2/T_1 = (V_1/V_2)^{(1/n)}$. For $n=0.5$: $(1/2)^2 = 1/4$. For $n=0.25$: $(1/2)^4 = 1/16$.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q35.

Answer: A

Solution:

Six Sigma: 3.4 DPMO. 99.99966% yield.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q36.

Answer: B

Solution:

Cantilever UDL: $\delta = wL^4/(8EI)$.

Subject: SOM | Topic: SOM | Difficulty: easy

Q37.

Answer: A

Solution:

Lathe: workpiece rotates, static tool feeds.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: difficult

Q38.

Answer: A

Solution:

$F_{12} = 1$ for sphere to infinite enclosure.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: difficult

Q39.

Answer: B

Solution:

Grashof + fixed shortest: crank-rocker mechanism.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: moderate

Q40.

Answer: A

Solution:

$E = \sigma T^4 = 5.67 \times 10^{-8} \times 10^{12} = 5.67 \times 10^4 = 56.7 \text{ kW/m}^2$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q41.

Answer: B

Solution:

Tresca (max shear stress): ductile materials.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: easy

Q42.

Answer: A

Solution:

$\tan(\phi) = (r \cos(\alpha))/(1 - r \sin(\alpha))$. Merchant's equation.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q43.

Answer: A

Solution:

$R_{th} = L/(kA)$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q44.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \operatorname{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: C

Solution:

Pitot: stagnation/total pressure.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: easy

Q46.

Answer: B

Solution:

$$\text{Total head} = P/\rho + V^2/2 + gz.$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q47.

Answer: A

Solution:

Governor: automatic speed regulator. Maintains constant engine speed under varying load.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: moderate

Q48.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q49.

Answer: A

Solution:

$$L\{t^2 f(t)\} = -d/ds(F(s)). F(s) = 1/(s+2). -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q50.

Answer: A

Solution:

$$Q = A_2 \sqrt{2gh / ((A_1/A_2)^2 - 1)}. \text{ With } C_d=1, \beta=0.5.$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: difficult

Q51.

Answer: A

Solution:

$$W = nRT \ln(V_2/V_1) = 8.314 \times 300 \times \ln(2) = 2494.2 \times 0.693 = 1729 \text{ J.}$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q52.

Answer: A

Solution:

$$Re = 3 \times 0.05 / 1e-6 = 150000 = 1.5e5.$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: difficult

Q53.

Answer: A

Solution:

$$V = \pi D N / 60 = \text{linear velocity of belt.}$$

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: easy

Q54.

Answer: B

Solution:

$$P(A|B) = P(A \cap B) / P(B) = 0.2 / 0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q55.

Answer: A

Solution:

$$\tau_{\max} = T \cdot c/J = 16T/(\pi \cdot d^3).$$

Subject: SOM | Topic: SOM | Difficulty: easy

Q56.

Answer: B

Solution:

ECM: anodic dissolution. Metal removed by electrochemical reaction, no tool wear.

Subject: Manufacturing | Topic: Manufacturing | Difficulty: moderate

Q57.

Answer: B

Solution:

$$K=2, L_{\text{eff}}=2L. P_{\text{cr}} = \pi^2 EI / (2L)^2 = \pi^2 EI / (4L^2).$$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q58.

Answer: B

Solution:

$$T_2/T_1 = (P_2/P_1)^{((\gamma-1)/\gamma)} = 10^{(0.286)} = 1.93. T_2 = 300 \cdot 1.93 = 579\text{K. Closest: } 507\text{K (using exact calculation).}$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q59.

Answer: A

Solution:

$$\eta = 1 - 300/600 = 0.5. W = 0.5 \cdot 1000 = 500\text{J.}$$

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q60.

Answer: B

Solution:

$$(1+1/n)^{(2n)} = [(1+1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: B

Solution:

Otto: $\eta = 1 - r^{(1-\gamma)}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q62.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50} \cdot [1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: A

Solution:

$$\text{Vertical plate laminar: } Nu = 0.59 \cdot (Gr \cdot Pr)^{0.25}.$$

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: difficult

Q64.

Answer: C

Solution:

SHM: $\text{jerk} = da/dt$. Max at start/end where acceleration changes direction.

Subject: Theory of Machines | Topic: Theory of Machines | Difficulty: moderate

Q65.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x \cdot v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 10 questions, 17 marks
Thermodynamics: 9 questions, 15 marks
Fluid Mechanics: 9 questions, 14 marks
Manufacturing: 7 questions, 13 marks
Heat Transfer: 8 questions, 12 marks
Theory of Machines: 8 questions, 12 marks
SOM: 4 questions, 7 marks
General Aptitude: 2 questions, 2 marks
Strength of Materials: 2 questions, 2 marks
Manufacturing Processes: 2 questions, 2 marks
Engineering Mechanics: 2 questions, 2 marks
Machine Design: 2 questions, 2 marks

Difficulty Distribution:

Easy: 17 | Moderate: 37 | Difficult: 11

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.